BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 015 · Bubble Sort and Insertion Sort

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Does linear search require a sorted list?

Q2. What happens in the fetch stage?
A. Work out its meaning
B. Carry out the instruction
C. Get the next instruction
D. Save a backup copy

Q3. What happens during the execute stage?

Q4. A CPU compares scores and adds a bonus. Correct both claims. Explain why. (a) The Control Unit stores documents forever. (b) The ALU stores the next instruction's address.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

bubble sort — Repeatedly compares and swaps adjacent out-of-order values.

insertion sort — Inserts each next item into a growing sorted section.

pass — One traversal of adjacent values in a sort.

key — Current value being inserted into a sorted section.

New learning

Bubble sort: Compare adjacent values and swap those in the wrong order; repeat passes until no swaps are needed.

After each pass: For ascending bubble sort, the largest remaining value moves towards the end on each full pass.

Insertion sort: Treat the first item as a sorted section; take the next key item and insert it into the correct place.

Keep a trace: Show each swap or insertion rather than jumping straight to the final sorted result.

Worked example

Show one full bubble-sort pass through [5, 1, 4, 2].

1. Compare 5 and 1: swap → [1, 5, 4, 2].

2. Compare 5 and 4: swap → [1, 4, 5, 2].

3. Compare 5 and 2: swap → [1, 4, 2, 5].

4. End of first pass: 5 has reached its final position.

5. More passes are needed to sort the remaining values.

We Do · Practise with guidance

Guided 1

Perform one bubble-sort pass on [4, 2, 3].

Worked method: Swap 4/2 → [2, 4, 3]; swap 4/3 → [2, 3, 4].

Guided 2

For insertion sort on [6, 2, 5], show what happens when inserting 2 into the sorted section.

Worked method: Begin [6]; insert 2 before 6 → [2, 6, 5].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. No; it examines items in sequence until a match is found or the list ends.

Q2. C — Get the next instruction

Q3. The operation specified by the decoded instruction is carried out.

Q4. (a) The Control Unit decodes instructions and controls CPU operations. (b) The ALU handles arithmetic and logic. Registers store data or addresses briefly.

I Do / We Do pages · Guided practice

Q1. Swap 4/2 → [2, 4, 3]; swap 4/3 → [2, 3, 4].

Q2. Begin [6]; insert 2 before 6 → [2, 6, 5].

Q3. Insert 5 between 2 and 6 → [2, 5, 6].

You Do pages · Code First

```
firstMB = 4
secondMB = 2
print(firstMB + secondMB)
```

Expected result: 6

You Do pages · Independent questions

Q1. Adjacent (neighbouring) values [1].

Q2. [1, 3, 2] [1]; [1, 2, 3] [1]; largest item reaches end [1].

Q3. First item 4 is initial sorted section [1]; insert 1 before 4 → [1, 4, 3] [1]; take key 3 [1]; insert between 1 and 4 → [1, 3, 4] [1].

Exit Ticket · Model answer

False

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

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BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

Guided 3

Finish insertion sort on [2, 6, 5].

Worked method: Insert 5 between 2 and 6 → [2, 5, 6].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Predict the combined file size.

```
firstMB = 4
secondMB = 2
print(firstMB + secondMB)
```

predict

Question 1

State what bubble sort compares. [1 mark]

Question 2

Show one bubble-sort pass on [3, 1, 2], giving the array after each swap. [3 marks]

Question 3

Explain how insertion sort orders the list [4, 1, 3]. [4 marks]

Exit Ticket · Answer before checking

One bubble-sort pass always completely sorts any list.

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Do Now Recall previous learning | I Do Read new knowledge | We Do Practise with guidance | You Do Work independently | Review & Improve Check understanding